

International Seminar on Revitalising Indigenous Knowledge Systems · ICSSR · 11–12 September 2026

Sub-theme: Digital Preservation, Artificial Intelligence, and Documentation of Indigenous Knowledge

An IKS-Grounded Multimodal Advisory System

Connecting Image-Based Plant Diagnosis with Classical Indian Agricultural Texts

वृक्षायुर्वेद · बृहत्संहिता · कृषिपराशर · उपवनविनोद · काश्यपीयकृषिसूक्ति · विश्ववल्लभ

Ankit Pawar · Dr. Akshay Pandey (Supervisor)

Department of Computer Science & Engineering, PDPM Indian Institute of Information Technology, Design and Manufacturing, Jaba Ipur

Centuries of plant-care knowledge — locked on the shelf



A farmer sees
dark, scabby patches
on apple leaves.

Where does advice come from today?

- A chemical dealer
- An AI chatbot ↓

THE HALLUCINATION PROBLEM

Chatbots answer confidently even when they do not know — and invent facts and sources. For indigenous knowledge that few people can verify, that is dangerous.

Six classical treatises hold the knowledge...

Vriksha-
yurveda

Brihat
Samhita

Krishi
Parashara

Upavana-
vinoda

Kashyapiya
Krishisukti

Vishva-
vallabha

...but three locks keep it on the shelf

1 Language

Sanskrit, or rare scattered translations

2 No digital form

Nothing structured or searchable

3 Symptom, not name

The texts say “dark scabby patches spreading” — never “apple scab”.
Modern labels do not exist in them.

THE SOURCES, IN ONE LINE EACH

The six texts — what each one contains

Vrikshayurveda

Surapala, c. 10th c.

The science of plant life — planting, nourishing, and treating sick trees

Brihat Samhita

Varahamihira, 6th c.

Encyclopaedia — rain, finding underground water, crop growth, tree treatment

Krishi Parashara

Parashara

The farming almanac — rain forecasting, sowing calendar, seed storage

Upavanavinoda

Sarngadhara

Garden and orchard care — planting, grafting, watering, protection

Kashyapiya Krishisukti

Kashyapa, before 800 CE

Cultivation from land to harvest — land, water, seeds, sowing, grain, vegetables

Vishvavallabha

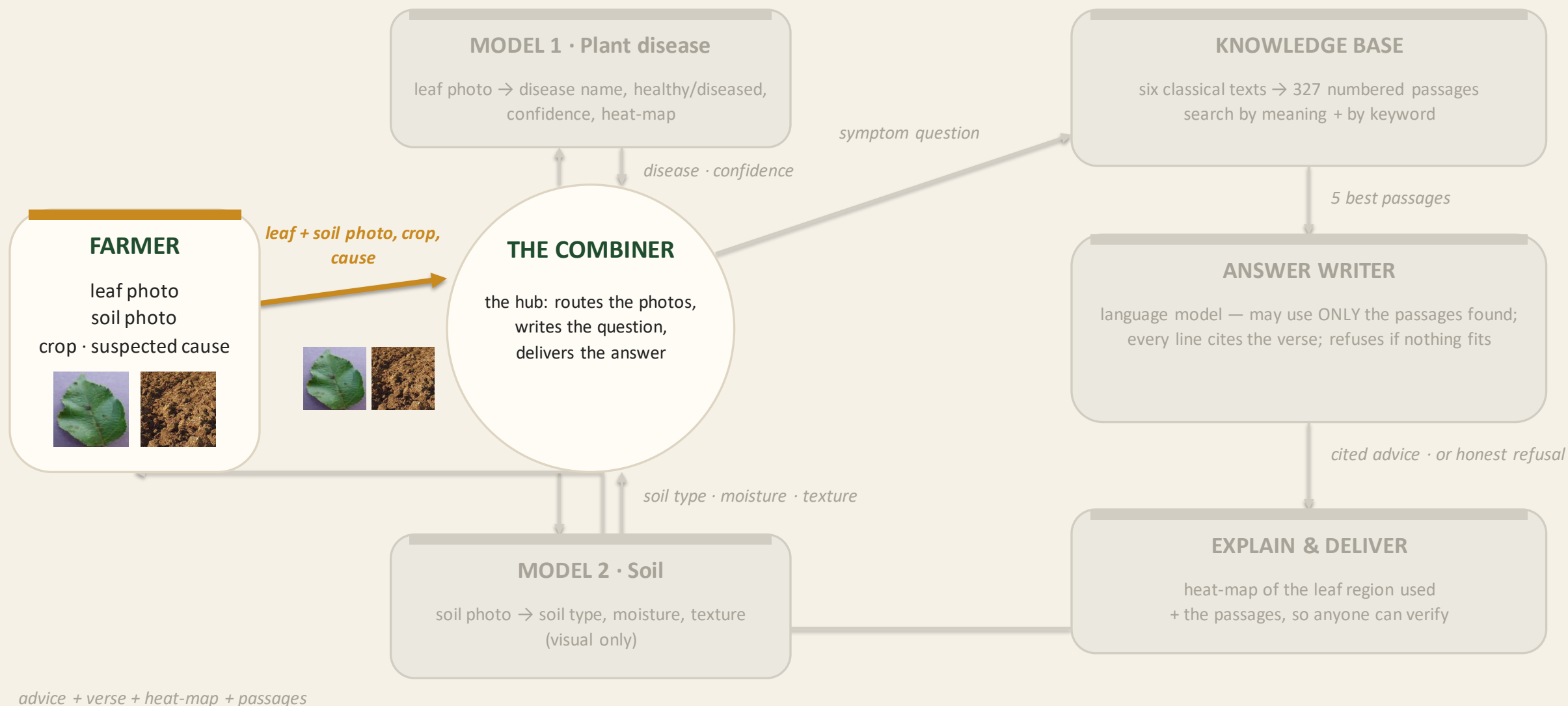
Chakrapani Mishra, 1577

Water-finding, horticulture, plant disease and pests — being added next

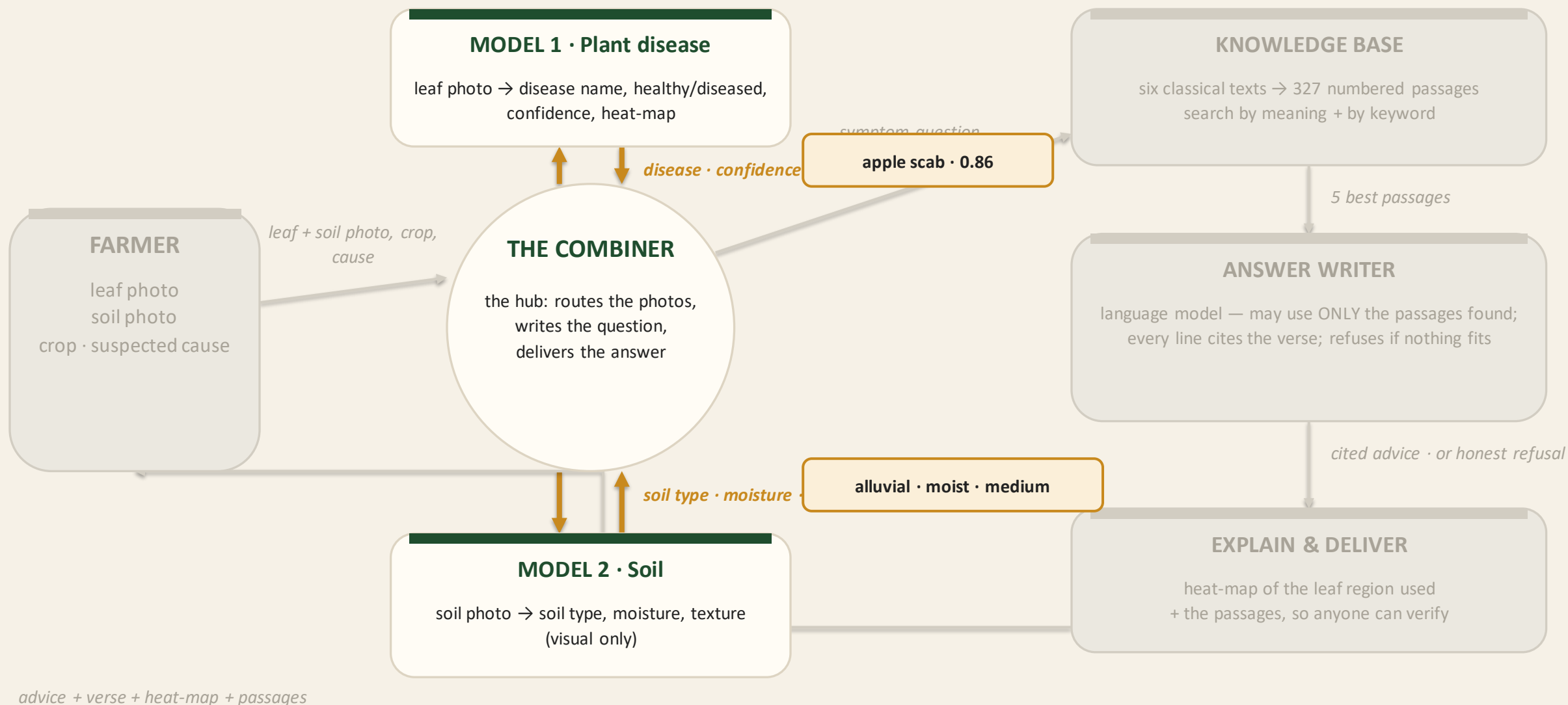
ONE REAL LINE, AS PUBLISHED

Vrikshayurveda v.193 — “To remove insects from the roots and branches of the trees, wise men should water the trees with cold water for seven days.” → symptom, then remedy, numbered: that is what makes the texts searchable.

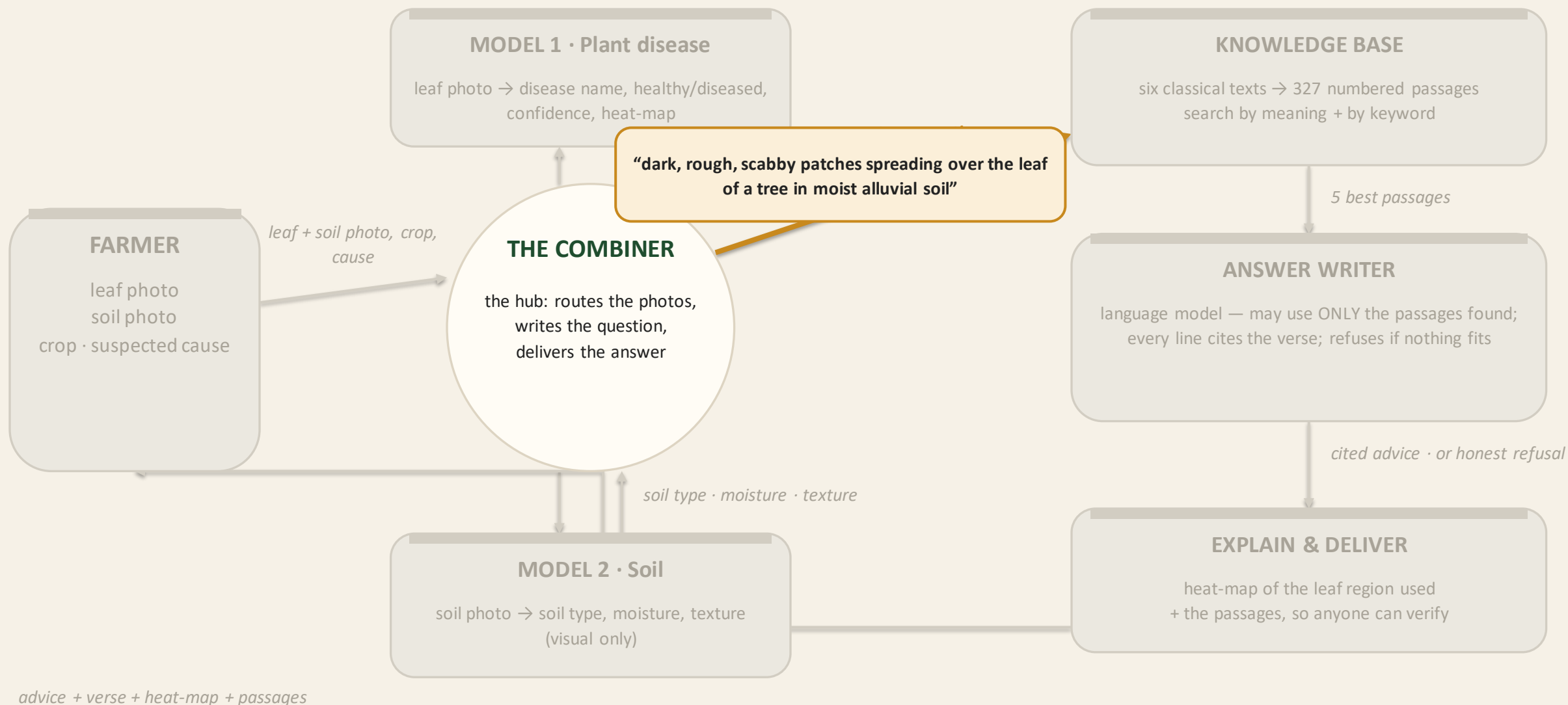
How it works — from photo to cited advice



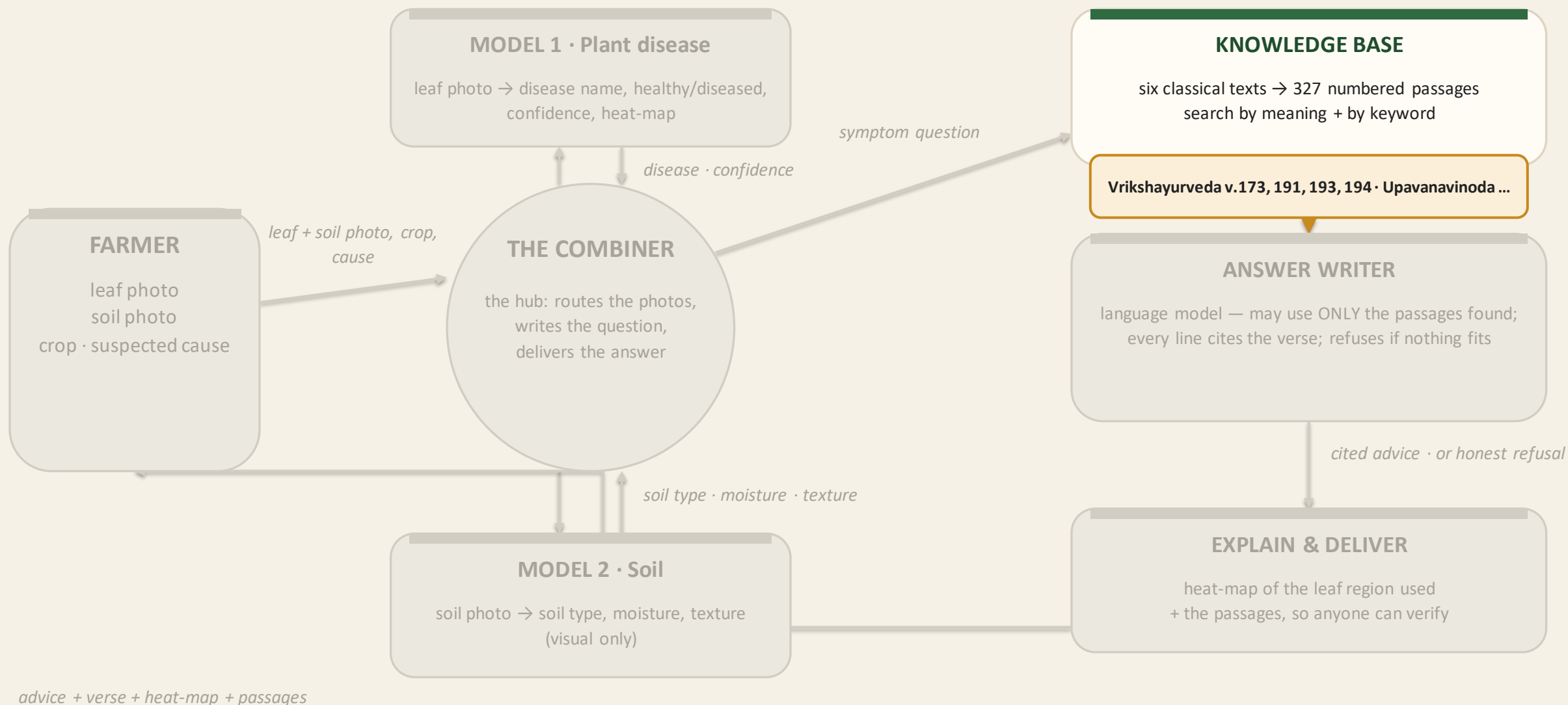
How it works — from photo to cited advice



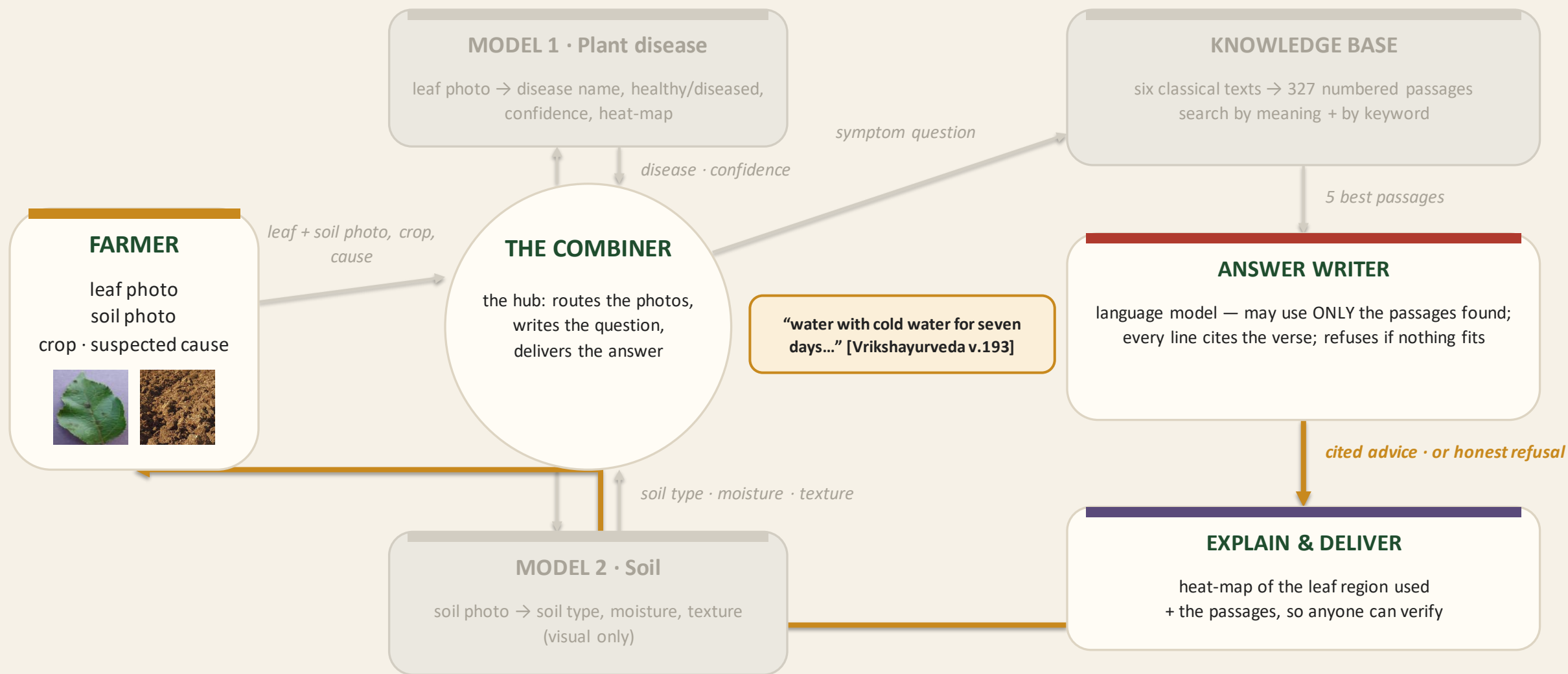
How it works — from photo to cited advice



How it works — from photo to cited advice



How it works — from photo to cited advice



Step 5 of 5 - advice is written only from those passages, cited, and returned with the heat-map

The contribution: search by symptom, not by name

SEARCH WITH THE MODERN LABEL

“Apple scab”

→ the texts return almost nothing. The phrase does not exist in them.

relevance 0.01 – 0.04

RE-DESCRIBE AS THE SYMPTOM

“dark, rough, scabby patches spreading over the leaf”

→ strong matches — the texts speak this language.

relevance 0.59 – 0.96

How the rewriting is done

A language model is given the diagnosis and asked to describe the visible symptoms in plain words — the way a cultivator would describe them. The crop becomes context; the symptom leads.

What this unlocks

- Searching by meaning beats searching by keywords: 0.94 vs 0.70
- A symptom carries across plants → cautious advice even for untrained crops, with a confidence figure
- Matches how the texts themselves prescribe — by symptom, not by crop

What the farmer sees — the interface

IKS Agricultural Advisory · prototype

INPUTS

Leaf photo



Soil photo



Crop

Apple ▾

Suspected cause (optional)

Not sure ▾

Get advice

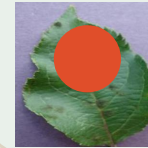
RESULT

Diagnosis

Apple — scab · confidence 0.86

Soil: alluvial · moist · medium texture

Where the model looked



heat-map on the lesion — not the background

Recommendation (from the texts only)

Cut away the affected parts and sprinkle the spots with water mixed with milk; feed the roots with kunapa water and cow-dung paste. Water with cold water for seven days.

Source: [Vrikshayurveda, v. 193–194] · [Vrikshayurveda, v. 191]

Retrieved passages (so anyone can verify)

v.193 “To remove insects from the roots and branches ... water the trees with cold water for seven days.”

v.194 “The worms can be overcome by the paste of milk, kunapa water, and cow dung ...”

Illustrative screen with real corpus passages. If the texts are silent, the result panel states: “The classical texts do not contain enough information for this case.”

Trust by design — four guarantees against fabrication

Cites the verse

Every recommendation is composed only from retrieved passages.

[Vrikshayurveda, v. 193]

Refuses when silent

If the texts do not cover a case, it says so instead of inventing.

100% honest refusal
0% fabricated citations

Attention verified

The heat-map must sit on the lesion. Early on it sat on the background of field photos — we retrained until it did not.

Human in the loop

Uncertain or out-of-scope cases are collected for an agronomist to review.

No silent self-learning from unverified guesses

0.01–0.04 → 0.59–0.96

relevance: modern label vs symptom rewriting

100% · 0%

honest refusal · fabricated citations

0.94 vs 0.70

search by meaning vs by keyword

For a plant the model was never trained on, it still advises — with a confidence figure and an explicit caution. Always “confidence”, never “accuracy”.

A text that is cited every day is preserved — and alive.

- Documentation and use reinforce each other — retrieval is preservation in practice
- Verifiable, not distorted — every answer traces to its source verse
- In the farmer's hands, not only in archives — a template for AI on Indigenous knowledge beyond agriculture

NEXT STEPS

- Validation with agronomists · add Vishvavallabha · expert-built evaluation set

Thank you

Backup — the flow as a loop

